

Title of Topic: NORMALIZATION

Course Code: 23IT201

Course Title: DataBase Management Systems

MODULE –II

Academic Year: 2024 - 2025 (Even Sem)

Module 2 – CONSTRAINTS AND SQL COMMANDS

DDL Commands - Create, Drop, Alter, Truncate, Rename, - DML Commands –

Normalization - 1NF, 2NF, 3NF, BCNF, 4NF -

Select, Insert, Update, Delete, Any, All, In, Exists, Non Exists, Union,
Intersection, Subqueries - nested, correlated, Joins- Inner, Outer, and Equi,
Functions - SUM, COUNT, AVG, MIN, MAX, Clauses - Group By, Having By,
Embedded SQL, Dynamic SQL, Transaction Concepts – Transaction model –
ACID Properties – Serializability

Data Normalization

- Formal process of decomposing relations with anomalies to produce smaller, *well- structured* and *stable* relations
- Primarily a tool to validate and improve a logical design so that it satisfies certain constraints that *avoid unnecessary duplication of data*



Well-Structured Relations

- A relation that contains minimal data redundancy and allows users to insert, delete, and update rows without causing data inconsistencies
- Goal is to avoid (*minimize*) anomalies
 - **Insertion Anomaly** – adding new rows forces user to create duplicate data
 - **Deletion Anomaly** – deleting a row may cause loss of other data representing completely different facts
 - **Modification Anomaly** – changing data in a row forces changes to other rows because of duplication

General rule of thumb: a table should not pertain to more than one entity type

EMPLOYEE2

<u>EmpID</u>	Name	DeptName	Salary	<u>CourseTitle</u>	DateCompleted
100	Margaret Simpson	Marketing	48,000	SPSS	6/19/201X
100	Margaret Simpson	Marketing	48,000	Surveys	10/7/201X
140	Alan Beeton	Accounting	52,000	Tax Acc	12/8/201X
110	Chris Lucero	Info Systems	43,000	Visual Basic	1/12/201X
110	Chris Lucero	Info Systems	43,000	C++	4/22/201X
190	Lorenzo Davis	Finance	55,000		
150	Susan Martin	Marketing	42,000	SPSS	6/19/201X
150	Susan Martin	Marketing	42,000	Java	8/12/201X

Question – Is this a relation?

Answer – Yes: unique rows and no multivalued attributes

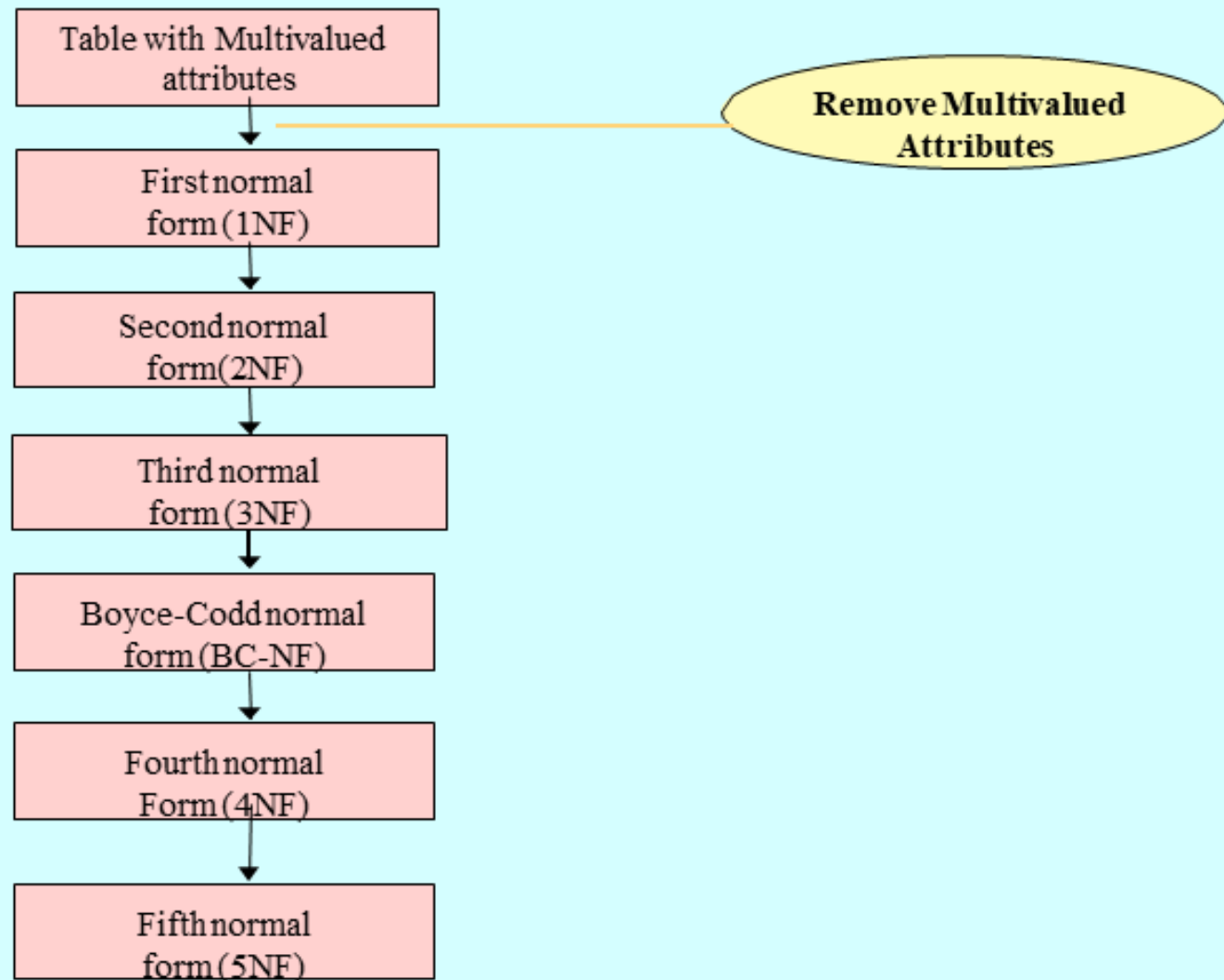
Question – What's the primary key?

Answer – Composite: EmpID, CourseTitle

Anomalies in this Table

- **Insertion** – can't enter a new employee without having the employee take a class
- **Deletion** – if we remove employee 140, we lose information about the existence of a Tax Acc class
- **Modification** – giving a salary increase to employee 100 forces us to update multiple records

Figure: 4-22 Steps in Normalization



First Normal Form (1NF)

- A relation is in first Normal Form if and only if all underlying domains contain atomic values only
- In other words, a relation doesn't have multivalued attributes (MVA)
- A primary key has been identified
- ***Every relation is in 1NF*** by definition
- 1NF example:

Student

<u>StudentId</u>	StuName	<u>CourseId</u>	CourseName	Grade
100	Mike	112	C++	A
100	Mike	111	Java	B
101	Susan	222	Database	A
140	Lorenzo	224	Graphics	B

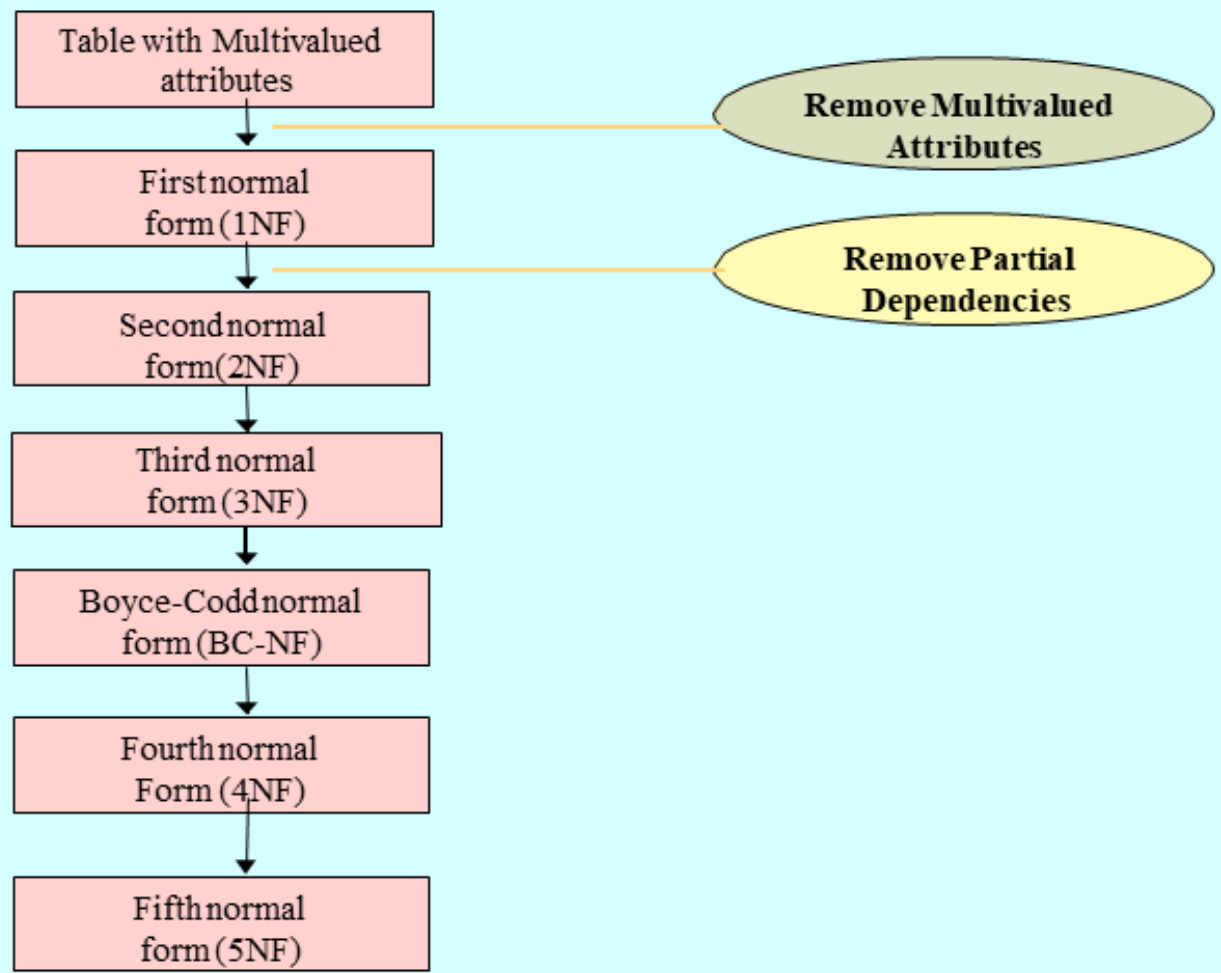


Course	Content
Programming	Java, c++
Web	HTML, PHP, ASP

We re-arrange the relation *table* as below, to convert it to First Normal Form.

Course	Content
Programming	Java
Programming	c++
Web	HTML
Web	PHP
Web	ASP

Figure: 4-22 Steps in Normalization

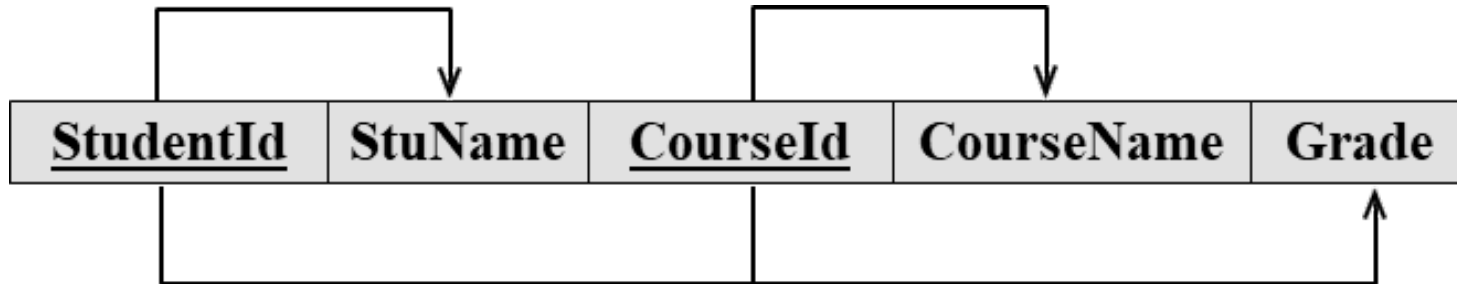


Second Normal Form (2NF)

- 1NF PLUS ***every non-key attribute is fully functionally dependent on the ENTIRE primary key***
 - Every non-key attribute must be defined by the entire key, not by only part of the key
 - No partial functional dependencies



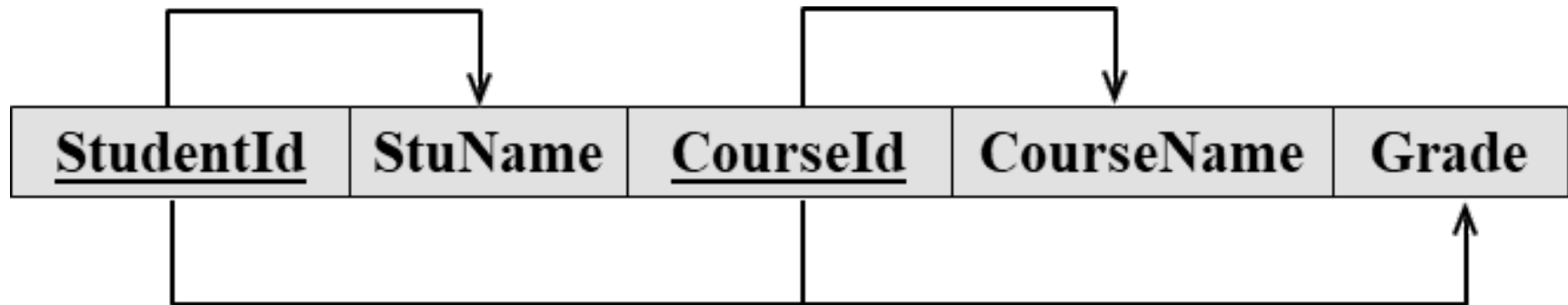
Functional Dependencies in Student



Can represent FDs with arrows as above, or

- StudentId → StuName,
- CourseId → CourseName
- StudentId, CourseId → Grade (and StuName, CourseName)

Any partial FDs ?



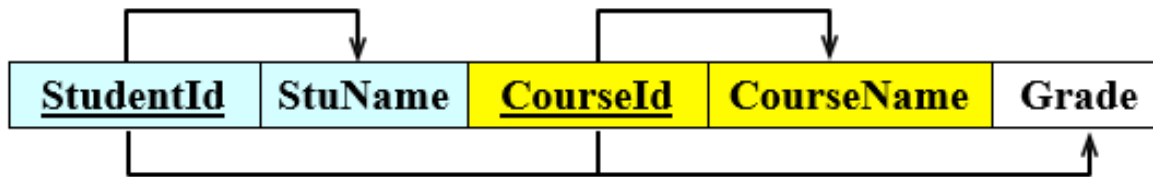
Can represent FDs with arrows as above, or

- StudentId $\rightarrow\rightarrow$ StuName,
- CourseId $\rightarrow\rightarrow$ CourseName
- StudentId, CourseId $\rightarrow\rightarrow$ Grade (and StuName, CourseName)

Therefore, NOT in 2nd Normal Form!!

2NF: Normalizing

To convert the partial dependencies into normal ones we have to breaking into more tables.



arrows mean functional dependency, below they mean FK constraints

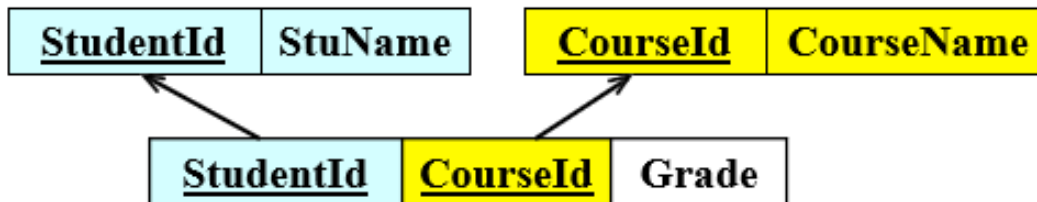
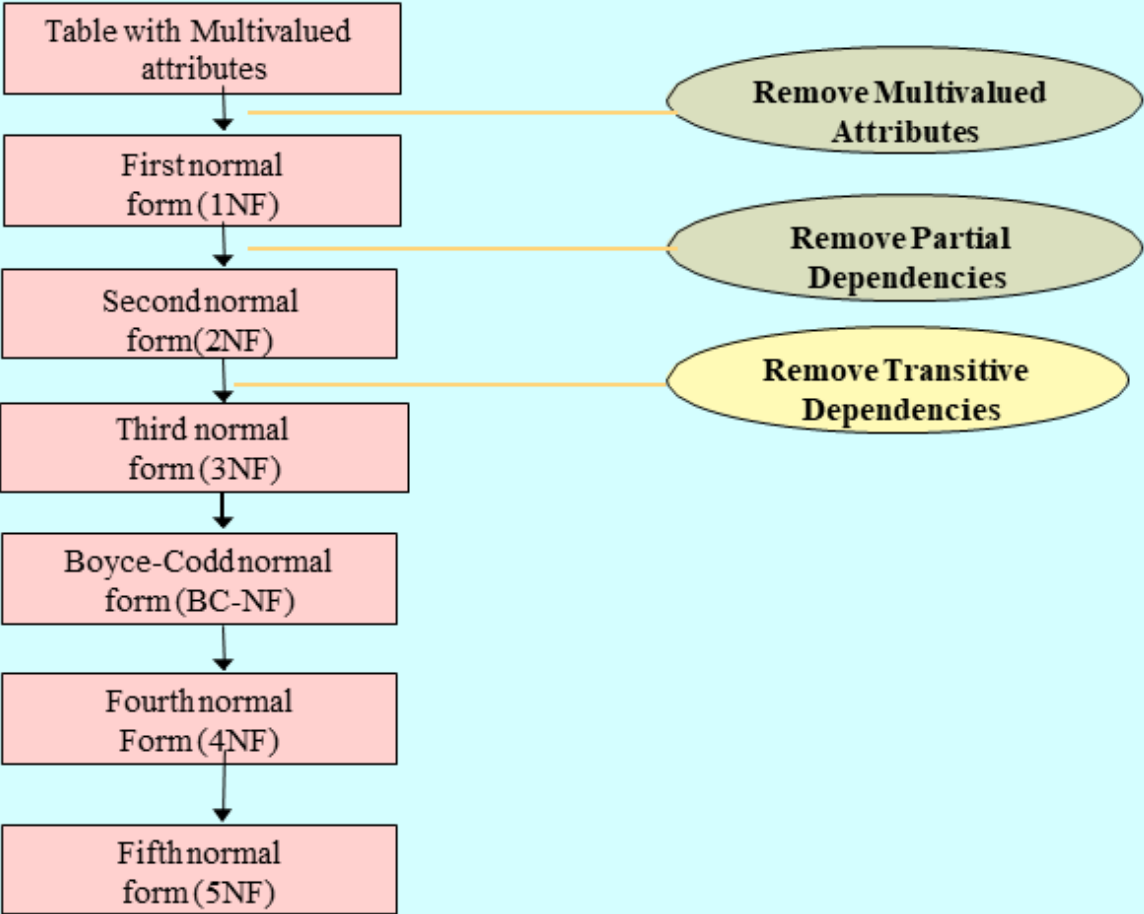


Figure: 4-22 Steps in Normalization

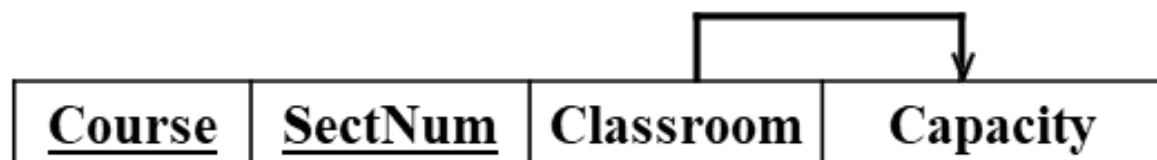


Third Normal Form

- 2NF and no **transitive dependencies**
- A **transitive dependency** is when a non-key attribute depends on another non-key attribute
- Note: This is called transitive, because the primary key is a determinant for another attribute, which in turn is a determinant for a third attribute

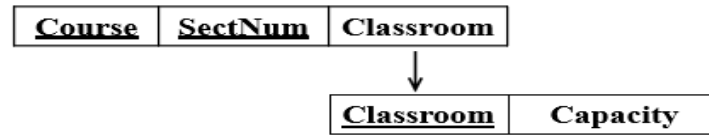


3NF Example



- Classroom $\twoheadrightarrow$ Capacity TRANSITIVE
- Any partial FDs? NO
- Any transitive FDs? YES !
 - How do we eliminate it?
 - By breaking into its own table

3NF Normalization



No non-prime attribute is transitively dependent on prime key attribute. For any non-trivial functional dependency,

$X \rightarrow A$, then

either – X is a superkey or,

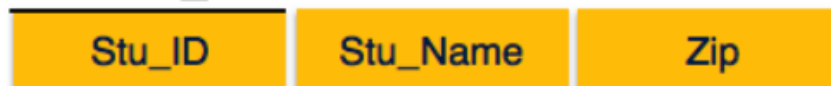
A is prime attribute.

Student_Detail



In the above Student_detail relation, Stu_ID is the key and only prime key attribute. We find that City can be identified by Stu_ID as well as Zip itself. Neither Zip is a super key nor is City a prime attribute. Additionally, $\text{Stu_ID} \rightarrow \text{Zip} \rightarrow \text{City}$, so there exists **transitive dependency**. To bring this relation into third normal form, we break the relation into two relations as follows –

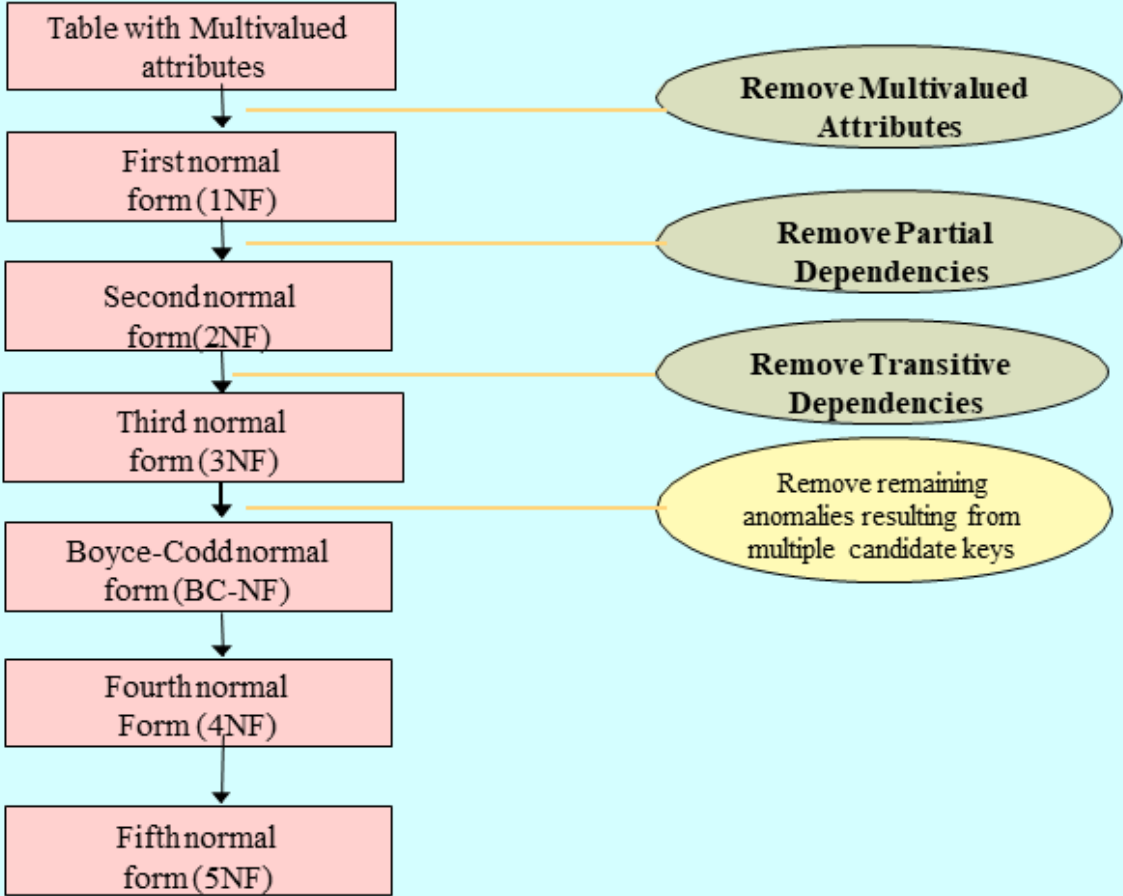
Student_Detail



ZipCodes



Figure: 4-22 Steps in Normalization



Boyce-Codd Normal form (BCNF)

- Slight difference with 3NF
- To be in 3NF but not in BNF, needs two composite candidate keys, with one attribute of one key depending on one attribute of the other
- Not very common.
- If a table contains only one candidate key, the 3NF and the BCNF are equivalent.



BCNF Example

Assume that

- For each subject, each student is taught by one Instructor
- Each Instructor teaches only one subject
- Each subject is taught by several Instructors

Course, Student → Instructor
Instructor → Course

Course, Student → Instructor
Instructor → Course

Course	Instructor	Student
CS 121	Dr. A. James	Bill Payne
CS 121	Dr. A. James	Tony Perez
CS 121	Dr. A. James	James Atkinson
CS 121	Dr. A. James	Linda Lee

Course	Instructor	Student
CS 141	Dr. T. Watson	Linda Lee
CS 141	Dr. T. Watson	Judith San
CS 141	Dr. T. Watson	Bill Jones
CS 141	Dr. P. Hold	Bill Payne
CS 141	Dr. P. Hold	A. White

Course	Instructor	Student
CS 101	Dr. M. Jones	Linda Lee
CS 101	Dr. M. Jones	Tony Perez
CS 101	Dr. M. Jones	Bill Payne

BCNF: Decompose into (Instructor, Course) and (Student, Instructor)

BCNF

- **Boyce-Codd normal form (BCNF)**

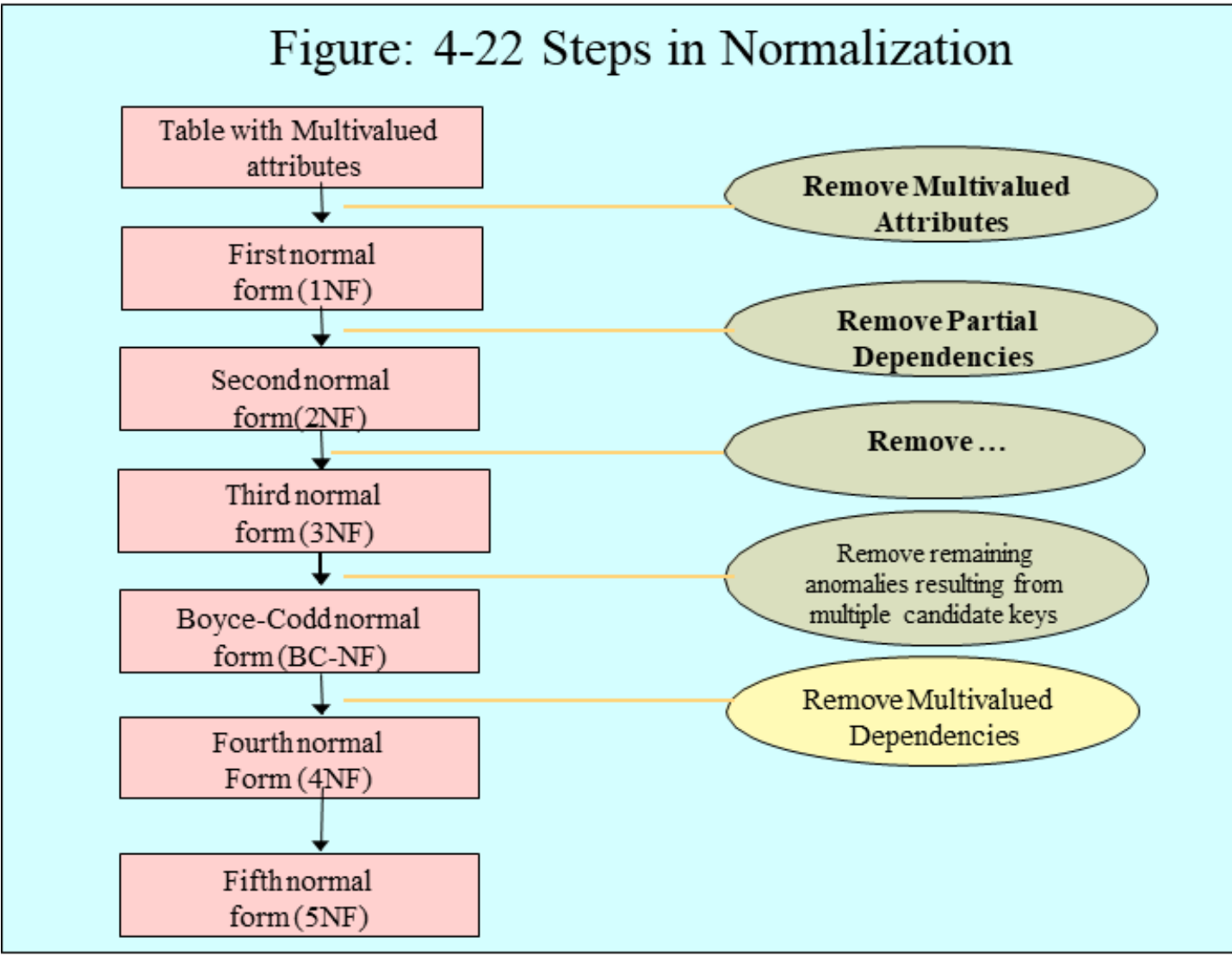
A relation is in BCNF, if and only if, every determinant is a candidate key.

- **The difference between 3NF and BCNF** is that for a functional dependency $A \twoheadrightarrow B$, 3NF allows this dependency in a relation if B is a primary-key attribute and A is not a candidate key,

whereas BCNF insists that for this dependency to remain in a relation, A must be a candidate key.



Figure: 4-22 Steps in Normalization



4NF

- **A multi-valued dependency exists when**
 - There are at least 3 attributes A, B, C in a relation and
 - For each value of A there is a well defined set of values for B, and a well defined set of values for C,
 - But the set of values for B is independent on the set of values for C
- **4NF = 3NF with no multi-valued dependency**



Assume that

- Each subject is taught by many Instructors
- The same books are used in many subjects
- Each Instructor uses a different book

Course	Instructor	Text
CS 121	Dr. A. James	Int to Com Science
CS 121	Dr. P. Hold	Comp Scien Int

Course	Instructor	Text
CS 141	Dr. T. Watson	Int to Com Science
CS 141	Dr. P. Hold	Comp Scien Int
CS 101	Dr. M. Jones	COMP SCIEN

Course, Instructor $\twoheadrightarrow$ Text

Course, Text $\twoheadrightarrow$ Instructor

4NF: Decompose into (Course, Instructor) and (Course, Text)

Example :INVOICE Data

Table with multivalued attributes, not in 1st normal form

<u>OrderID</u>	Order Date	Customer ID	Customer Name	Customer Address	<u>ProductID</u>	Product Description	Product Finish	Product StandardPrice	Ordered Quantity
1006	10/24/2010	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
					5	Writer's Desk	Cherry	325.00	2
					4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
					4	Entertainment Center	Natural Maple	650.00	3

INVOICE relation (1NF)

Table with no multivalued attributes and unique rows

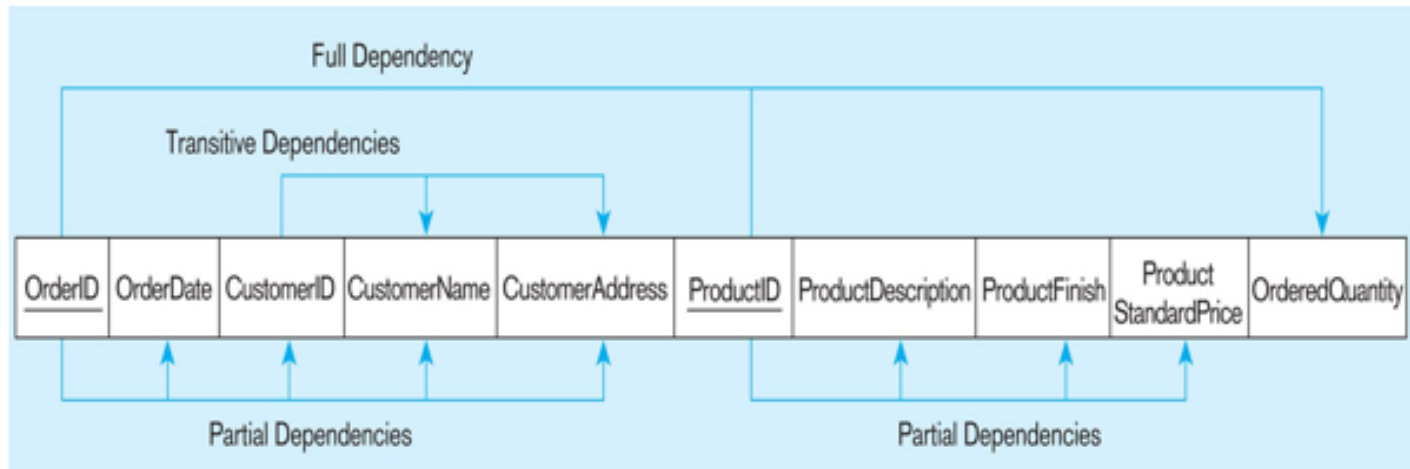
<u>OrderID</u>	Order Date	Customer ID	Customer Name	Customer Address	<u>ProductID</u>	Product Description	Product Finish	Product StandardPrice	Ordered Quantity
1006	10/24/2010	2	Value Furniture	Plano, TX	7	Dining Table	Natural Ash	800.00	2
1006	10/24/2010	2	Value Furniture	Plano, TX	5	Writer's Desk	Cherry	325.00	2
1006	10/24/2010	2	Value Furniture	Plano, TX	4	Entertainment Center	Natural Maple	650.00	1
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	11	4-Dr Dresser	Oak	500.00	4
1007	10/25/2010	6	Furniture Gallery	Boulder, CO	4	Entertainment Center	Natural Maple	650.00	3

this is relation, but not a well-structured one. WHY?

Anomalies in this Table

- **Insertion**—if new product is ordered for order 1007 of existing customer, customer data must be re-entered, causing duplication
- **Deletion**—if we delete the Dining Table from Order 1006, we lose information concerning this item's finish and price
- **Update**—changing the price of product ID 4 requires update in several records

Figure 4-27 Functional dependency diagram for INVOICE



Order_ID $\twoheadrightarrow$ Order_Date, Customer_ID, Customer_Name, Customer_Address

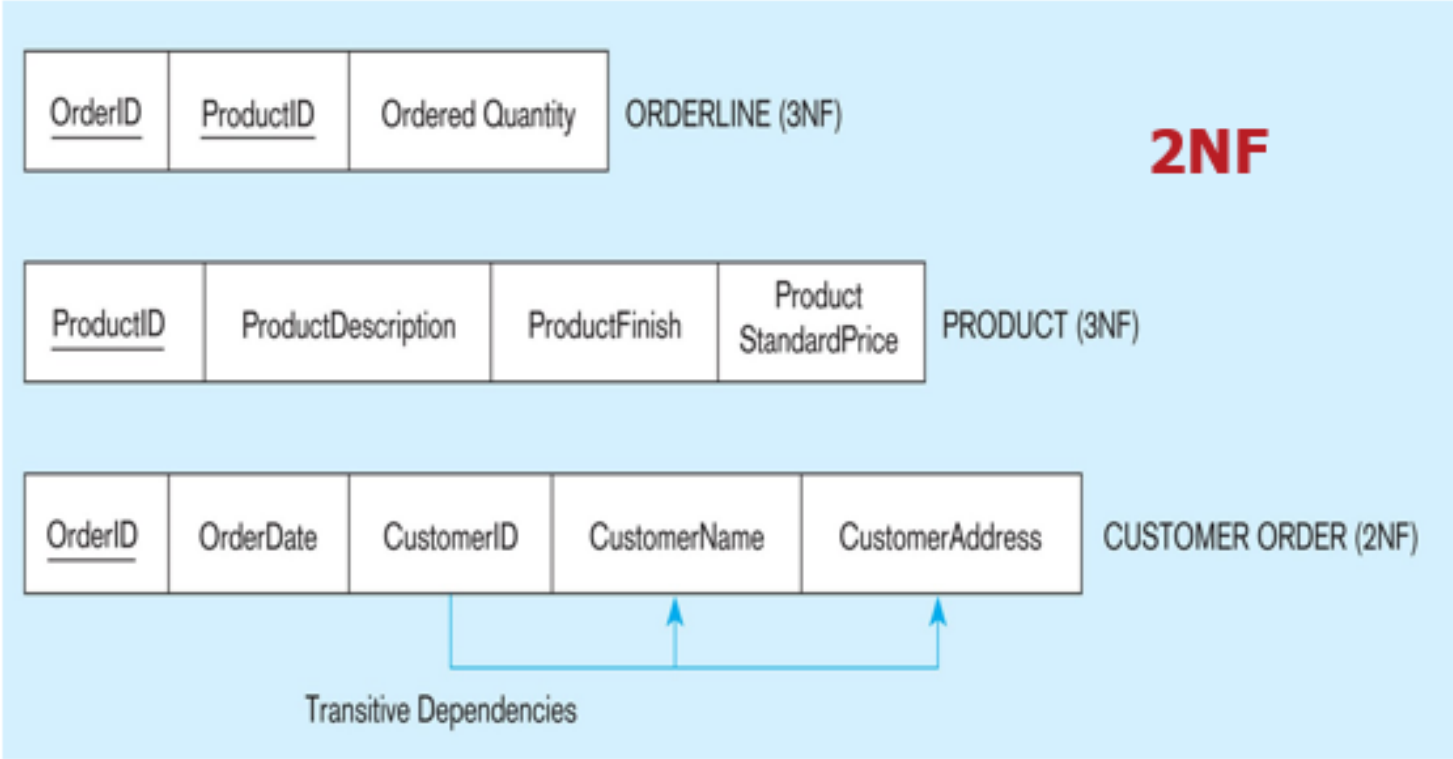
Customer_ID $\twoheadrightarrow$ Customer_Name, Customer_Address

Product_ID $\twoheadrightarrow$ Product_Description, Product_Finish, Unit_Price

Order_ID, Product_ID $\twoheadrightarrow$ Order_Quantity

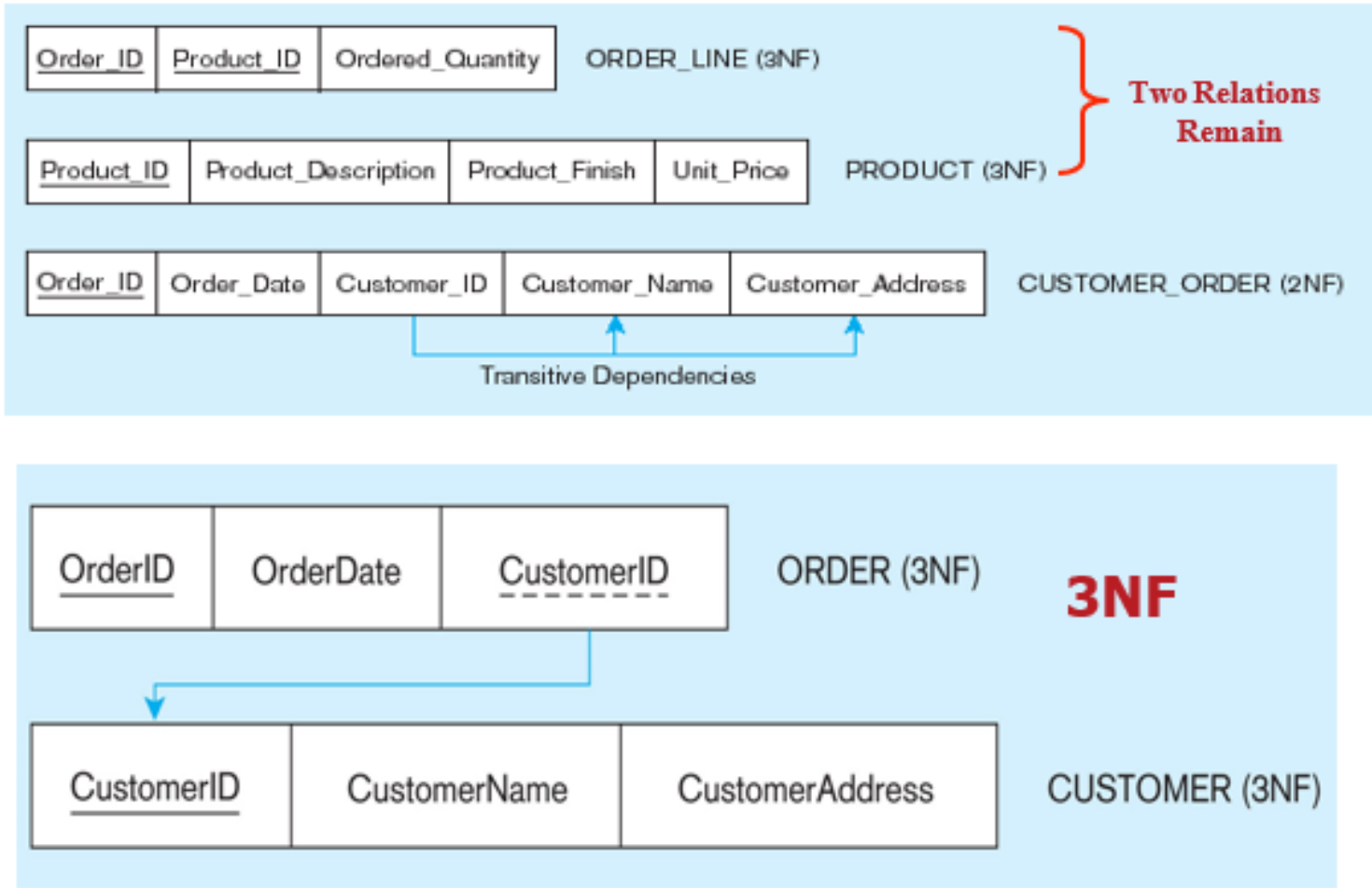
Therefore, NOT in 2nd Normal Form

Figure 4-28 Partial Dependencies were Removed (2NF)



Partial dependencies are removed, but there are still transitive dependencies

Figure 4-29 Transitive Dependencies were Removed (3NF)



Try yourself

TABLE 4-6 Parking Tickets at Millennium College

Parking Ticket Table

St ID	L Name	F Name	Phone No	St Lic	Lic No	Ticket #	Date	Code	Fine
38249	Brown	Thomas	111-7804	FL	BRY 123	15634	10/17/12	2	\$25
						16017	11/13/12	1	\$15
82453	Green	Sally	391-1689	AL	TRE 141	14987	10/05/12	3	\$100
						16293	11/18/12	1	\$15
						17892	12/13/12	2	\$25

- 1.Convert to 1NF Relation
- 2.Draw dependency diagram showing all functional dependencies
- 3.Identify anomalies
- 4.Convert to 3NF Relations
- 5.Develop EER Diagram with appropriate cardinalities